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Certification of Molecular Dynamics Trajectories with NMR Chemical Shifts

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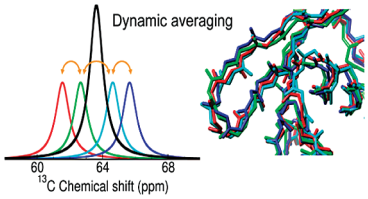
ABSTRACT Molecular dynamics ensembles of proteins generated by different force fields (AMBER ff99, ff99SB, ff03) have been quantitatively assessed based on their back-calculated Cα, Cβ, and C′ chemical shifts in comparison with NMR experiments. For the latest generation of force fields, a substantial improvement is found for ensemble-averaged chemical shifts over individual snapshots. Explicit inclusion of protein dynamics provides the largest improvement for Cβ chemical shifts, which are dominated by φ, ψ, and χ_1 dihedral angle distributions. Since NMR chemical shifts are available for a vast number of proteins, this novel strategy opens up the possibility to quantitatively certify molecular dynamics simulations on an unprecedented scale.

SECTION Macromolecules, Soft Matter



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The quantitative prediction of many protein properties at the atomic level critically depends on the accuracy of the underlying molecular mechanics force field.[1] Progress in force fields can be assessed by direct comparison between computed and experimental biophysical properties, whereby NMR spin relaxation parameters, residual dipolar couplings (RDCs), and scalar J-couplings, have proven particularly useful.[2−8] While backbone ^{15}N−^{1}H and methyl side-chain groups have been the most accessible parameters, validation with other protein parts, including many side chains,[9,10] has been limited. By contrast, NMR chemical shift information for both backbone and side-chain atoms is widely available through the BioMagResBank (BMRB).[11] Here, we demonstrate how chemical shift information permits the comprehensive assessment of the quality of molecular dynamics ensembles generated with different force fields.

Several approaches are now available for the increasingly quantitative prediction of chemical shifts from a 3D protein structure.[12−18] These include database-derived hypersurfaces in terms of dihedral angles and secondary structure in combination with analytical equations as implemented in the SHIFTX program[14] and the combination of local sequence homology with dihedral angle similarity as implemented in SPARTA.[15] Alternatively, structure versus chemical shift relationships have been parametrized based on quantum chemical calculations of small protein fragments and implemented in SHIFTS.[12,19]

Chemical shifts do not only depend on a protein's average (static) structure as they reflect also the presence of internal dynamics.[20] On the basis of this property, Berjanskii and Wishart derived relationships between secondary chemical shift changes and motional parameters.[21,22] Here, we use ^{13}C chemical shifts averaged over molecular dynamics trajectories to directly assess the accuracy of molecular dynamics (MD) simulations and their force fields. MD trajectories of calbindin D_{9k} and ubiquitin, which were performed using AMBER[23] with the AMBER ff99,[24] ff99SB,[25] and ff03[26] force fields in explicit SPC/E water at 300 K, show stable behavior over 30 ns to 1 μs lengths, as described previously.[2,10,27] Chemical shifts of all backbone Cα, Cβ, and carbonyl C′ atoms were computed for snapshots sampled every 1−10 ps using SHIFTS, SHIFTX, and SPARTA and compared with their experimental counterparts (BMRB entries 6457 and 16340). For calbindin, only atoms at a distance ≥8 Å from the bound Ca^{2+} ions were analyzed to keep ionic effects on chemical shifts small.

Figure 1 shows for different types of ^{13}C chemical shifts of ubiquitin the root-mean-square deviation (RMSD) between experiment and the prediction by SHIFTS for each snapshot as well as that for the calculated shifts averaged over the MD ensembles. Included in the figure are also the results for the X-ray crystal and NMR structures. The large widths of RMSD distributions for all three chemical shift types attest to the sensitivity of these NMR observables to structural variations displayed by the three room-temperature MD ensembles. Importantly, averaging of the predicted chemical shifts over the full ensembles substantially improves the agreement (by lowering the RMSD), as demonstrated in Figure 1.

The ff99SB and ff03 ensembles display better overall performance than the older ff99 ensemble for both ubiquitin (Figure 1) and calbindin (Table 1; Supporting Information), with the largest improvement found for the Cβ chemical shifts. These chemical shifts primarily reflect the backbone φ, ψ, and χ_1 dihedral angles,[12,14,15,28,29] indicating that the

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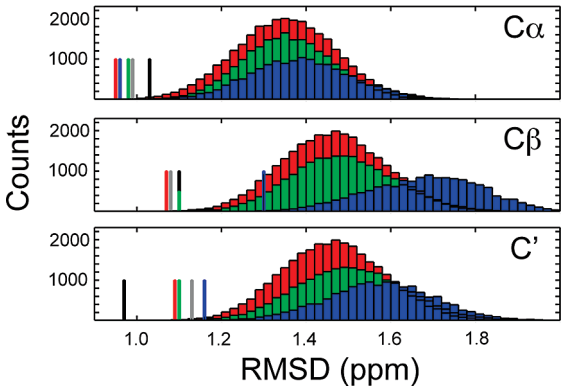

Figure 1. The RMSD of experimental and predicted chemical shifts via SHIFTS for Cα (top), Cβ (middle), and C′ (bottom) atoms of ubiquitin for 30 ns MD trajectories using different force fields, ff99SB (red), ff03 (green), and ff99 (blue). The corresponding ensemble averages are indicated as vertical lines of the same color. The black and gray lines belong to the X-ray crystal structure (1UBQ) and the NMR structure (1D3Z), respectively. For clarity, histograms for ff03 and ff99 are scaled along the y-axis by factors of $1/2^{1/2}$ and 1/2, respectively.

Table 1. RMSDs between Experimental and Calculated Chemical Shifts (in ppm) for Ubiquitin (Ubi) and Calbindin D_{9k}[a]

		ff99	99SB	99SB(1 μs)	ff03	PDB1[b]	PDB2[c]
	Cα	0.96	0.95	0.90	0.98	1.03	0.99
Ubi	Cβ	1.30	1.07	1.04	1.10	1.10	1.08
	C′	1.16	1.09	1.10	1.10	0.97	1.13
	Cα	1.39	1.17		1.14	1.52	1.39
D_{9k}	Cβ	1.08	0.92		0.83	1.29	1.31
	C′	1.28	1.09		1.12	1.90	1.40

[a] Calculated shifts are determined from the 30 ns MD ensembles as well as from individual static structures. [b] PDB entries 1UBQ for Ubi and 1B1G for calbindin D_{9k} were used for the prediction. [c] PDB entries 1D3Z for Ubi (first model) and 2BCA for calbindin D_{9k} were used for the prediction.

recent force fields provide a more realistic structural dynamic representation of these soft degrees of freedom.[4,25] Chemical shift averages of the ff99SB and ff03 ensembles show similarly good performance for both ubiquitin and calbindin (Table 1).

The dependence of the RMSD values of ubiquitin as a function of trajectory length between 0.1 ns and 1 μs is depicted in Figure 2. It demonstrates how increased sampling of conformational space steadily reduces the RMSD for the 3 ^{13}C-atom types with convergence setting in between 100 and 200 ns.

The MD ensemble-averaged chemical shifts for ff99SB and ff03 are for both proteins more accurate than the chemical shifts derived from experimental X-ray and NMR structures (Table 1). The only exception is the C′ chemical shift prediction from 1UBQ. The sensitivity of C′ chemical shifts to local geometry and hydrogen bonding involving the same peptide plane[12] indicates that in the MD ensembles further improvement of protein backbone structure, including hydrogen bonding, may be possible.

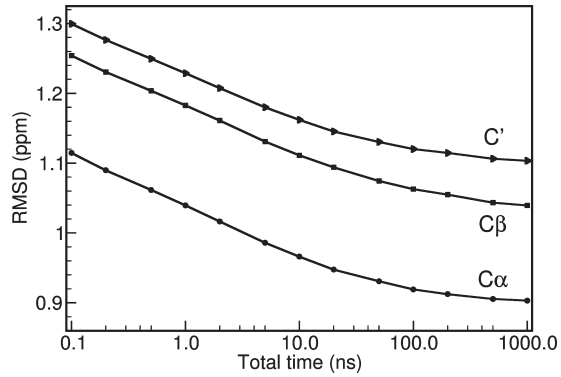

Figure 2. Dependence of the ^{13}C chemical shift RMSDs for ubiquitin as a function of trajectory length between 0.1 ns and 1 μs.

The SHIFTS method differs from the other two methods by being based on quantum chemical calculations of static structures. In principle, it permits determination of the absolute amount of dynamics present in solution. The SHIFTX and SPARTA methods, on the other hand, rely on empirical parametrizations using chemical shift database information of proteins experiencing internal dynamics. Some degree of motional averaging is thereby inherent in these parametrizations, analogous to Karplus relationships for ^{3}J-couplings.[7,30] Nonetheless, all three approaches produce similar results for both proteins tested for the MD ensembles and those structures that were directly used for parametrization, with SPARTA yielding the smallest RMSDs (Supporting Information). The methods are computationally quite efficient (1−6 s per MD snapshot) and can be easily parallelized when applied to ensembles.

All chemical shifts used here were directly taken from the BMRB without altering their referencing. Application of a uniform chemical shift offset that minimizes the RMSD leaves the results essentially unchanged (Supporting Information). For all MD simulations, including the 1 μs trajectory, the RMSDs significantly exceed the experimental chemical shift uncertainties.[11] Hence, there is clear potential for improvement for both force fields and chemical shift predictions by iteratively guiding advances of these methods using direct feedback from changes made to the force field or the chemical shift prediction. According to Figure 2, MD trajectories around 100 ns or longer, which start to be routine, should be most useful for this task. We anticipate that ongoing progress in the quantitative prediction of chemical shifts by quantum chemical and other methods will further increase their utility in the context of molecular modeling and dynamics studies.

For selected proteins, various types of NMR parameters have already proven valuable for cross-validation of MD ensembles, and the chemical shift analysis used here further corroborates these results. Moreover, for a vast and rapidly growing number of proteins studied by NMR (> 5000[11]), chemical shifts represent the main and often only source of NMR information. The powerful combination of dynamics simulations and NMR chemical shifts presented here thereby opens up the possibility to certify MD simulations on a very large scale.

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SUPPORTING INFORMATION AVAILABLE Methods section describing MD simulations and chemical shift RMSD calculations; 12 tables analogous to Table 1 for SHIFTX and SPARTA; ^{13}C-site-specific analysis based on the 1 μs trajectory of ubiquitin using SHIFTS, SHIFTX, and SPARTA (three figures and one table). This material is available free of charge via the Internet at http://pubs.acs.org.

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